

WHITEHORSE, YT, Canada

WMO: 719640

Lat: 60.7331N

Lon: 135.0978W

Elev: 706

StdP: 93.12

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-38.1	-33.5	-42.2	0.1	-38.1	-38.1	0.1	-33.4	12.6	-5.5	11.4	-6.3	0.7	0	0.762

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	25.8	14.3	23.5	13.3	21.6	12.6	15.1	23.8	14.1	22.0	13.2	20.2	3.4	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
11.6	9.3	16.3	10.5	8.6	15.8	9.6	8.1	15.3	44.3	24.0	41.6	22.1	39.1	20.4	18.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	10.4	9.4	8.4	DB	-40.0	28.8	3.5	2.0	-42.5	30.3	-44.5	31.5	-46.5	32.6	-49.0	34.1
(5)				WB	-40.1	16.4	3.4	1.0	-42.5	17.2	-44.6	17.8	-46.5	18.4	-49.0	19.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	0.1	-15.7	-11.8	-7.3	1.4	7.9	12.7	14.5	12.9	7.6	0.7	-9.0	-13.2
	DBStd	12.37	10.86	8.63	7.51	4.37	3.74	3.14	2.57	3.22	3.34	4.62	7.53	9.19
	HDD10.0	3968	796	611	535	259	87	7	1	8	86	289	569	721
	HDD18.3	6657	1054	844	793	509	323	172	122	172	323	547	819	979
	CDD10.0	362	0	0	0	0	22	89	141	97	13	1	0	0
	CDD18.3	9	0	0	0	0	0	3	3	3	0	0	0	0
	CDH23.3	214	0	0	0	0	7	68	73	67	0	0	0	0
	CDH26.7	40	0	0	0	0	1	17	11	12	0	0	0	0
Wind	WSAvg	3.7	3.7	3.8	3.7	3.7	3.6	3.3	3.2	3.3	3.8	4.0	3.9	4.1
Precipitation	PrecAvg	283	20	13	13	8	15	34	39	41	34	25	22	20
	PrecMax	394	60	33	50	27	42	92	112	110	68	55	56	43
	PrecMin	161	2	0	0	0	2	6	9	10	7	4	4	4
	PrecStd	49	10	8	10	7	11	19	21	21	16	11	11	10
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.7	4.4	9.4	17.3	24.1	28.9	28.0	28.3	19.7	13.1	5.6	5.5
		MCWB	2.6	1.4	3.8	8.5	11.9	14.7	14.7	15.7	11.2	7.8	2.7	2.2
	2%	DB	2.2	2.6	5.7	12.6	21.3	25.0	25.2	25.2	16.7	10.1	3.6	2.7
		MCWB	0.1	0.0	1.5	5.4	10.5	13.4	14.1	14.4	10.2	5.9	1.3	0.8
	5%	DB	0.4	0.9	3.8	10.0	18.4	22.5	23.2	22.3	14.8	8.2	1.9	0.7
		MCWB	-1.2	-1.3	0.2	3.7	9.1	12.4	13.6	13.5	9.0	4.6	0.0	-0.9
	10%	DB	-1.4	-0.7	2.1	8.1	16.0	20.2	21.2	19.8	13.2	6.6	0.4	-1.2
		MCWB	-2.7	-2.4	-1.0	2.6	7.9	11.4	13.0	12.2	8.2	3.6	-1.3	-2.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.4	1.7	4.0	8.5	12.7	15.9	16.4	16.2	12.4	8.6	3.1	2.6
		MCDB	5.2	4.0	8.8	17.0	22.6	26.8	24.7	26.7	17.8	12.1	5.2	5.1
	2%	WB	0.3	0.2	1.8	5.8	11.0	14.2	15.1	14.9	10.7	6.5	1.5	0.8
		MCDB	2.0	2.3	5.2	11.9	19.9	23.5	23.0	23.8	15.3	9.4	3.2	2.6
	5%	WB	-1.2	-1.2	0.4	4.0	9.6	13.1	14.2	13.7	9.6	4.9	0.1	-0.8
		MCDB	0.3	0.7	3.5	9.2	17.6	21.2	21.6	21.4	13.9	7.6	1.9	0.5
	10%	WB	-2.8	-2.4	-0.8	2.9	8.3	11.9	13.4	12.6	8.6	3.7	-1.3	-2.5
		MCDB	-1.4	-0.6	1.9	7.7	15.2	18.7	20.2	18.7	12.5	6.3	0.3	-1.3
Mean Daily Temperature Range	5% DB	MDBR	7.0	8.6	10.5	10.4	11.9	11.1	10.6	9.0	6.4	5.9	6.7	
		MCDBR	6.4	7.1	9.5	13.2	16.2	16.0	15.3	15.7	11.4	8.1	6.1	6.8
		MCWBR	5.3	5.6	6.3	7.7	8.0	7.2	6.8	7.1	6.4	5.3	5.0	5.8
	5% WB	MCDBR	6.5	6.8	8.8	12.2	14.6	14.4	13.2	14.2	10.0	7.1	5.9	6.4
		MCWBR	5.5	5.5	6.0	7.4	7.3	6.7	6.2	6.6	5.8	4.8	4.9	5.6
Clear-Sky Solar Irradiance	taub	0.235	0.252	0.267	0.334	0.363	0.365	0.357	0.348	0.304	0.272	0.213	0.185	
	taud	2.035	2.249	2.254	2.101	2.273	2.365	2.413	2.452	2.558	2.449	2.114	1.963	
	Ebn at Noon	499	759	869	852	858	863	862	842	833	737	545	410	
	Edh at Noon	42	64	89	128	117	110	102	91	68	53	37	27	
All-Sky Solar Radiation	RadAvg	0.35	1.17	2.78	4.64	5.51	5.57	5.08	3.99	2.44	1.19	0.42	0.18	
	RadStd	0.03	0.10	0.19	0.23	0.26	0.31	0.31	0.35	0.21	0.12	0.05	0.01	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	+2.13	+2.76	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (7 neighbors)	N/A	+1.42	+1.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page